Misr University for Science and Technology

College of Information Technology

Submitted by

Omar Ayman Abdelfattah	200033242	CS
Mariam Mohamed Ahmed	200026537	CS
Salma Samer Adel	200033188	CS
Nouran Mahmoud Elsayed	200026495	CS
Omar Wael Adawy	200023390	CS
Fares Hatem Habib	200021941	CS

Under Supervision of

.Dr. Eman Karam

CS401 – Seminar

Spring 2026

Table of Contents

1. Living Inside the Internet
2. What Cybersecurity Means
3. Core Goals of Cybersecurity
4. Cybersecurity in Daily Life
5. Common Cyber Threats
6. Main Categories of Cybersecurity
7. Specialized Branches and Operations
8. How Cybersecurity Works in Practice
9. Cybersecurity Careers
10. Impact on Society and Countries
11. Cyber Operations, Competition, and Ethics
12. Final Takeaways
13. References

This document expands the key ideas presented in the seminar slides into a structured report suitable for study, revision, and final submission.

1. Living Inside the Internet

Today, people do not simply use the internet as an external tool; they depend on it as an environment where daily life happens. Phones, laptops, cloud accounts, online classes, digital banking, e-commerce, remote work platforms, hospital records, smart TVs, and home devices all generate and exchange information continuously.

Each digital interaction creates value. A login can unlock personal identity, a payment can move money, and a cloud account can store years of private files. Because these actions carry real financial, social, and operational value, they also attract attackers. The more connected a person or institution becomes, the more important cyber protection becomes.

This explains why cybersecurity is no longer a niche topic for specialists only. It is now a basic condition for safe participation in modern life, similar to safety rules in transportation, medicine, and finance.

2. What Cybersecurity Means

Cybersecurity is the practice of protecting systems, networks, devices, applications, and data from unauthorized access, misuse, disruption, destruction, or theft. In practical terms, it protects both the technology itself and the trust people place in technology.

A complete cybersecurity program does more than prevent attacks. It also tries to detect threats early, respond when incidents happen, limit damage, restore operations, and learn from events so defenses improve over time.

This broader understanding is important because no organization can promise perfect prevention. Strong security is built on readiness, visibility, and recovery as much as on blocking malicious activity.

3. Core Goals of Cybersecurity

Cybersecurity is often explained through four foundational goals. Confidentiality means information is accessible only to authorized people. Integrity means data remains correct, trustworthy, and protected from unauthorized modification. Availability means services and systems stay accessible when needed. Resilience goes further by ensuring organizations can continue operating and recover quickly even during disruption.

These goals help organizations make balanced decisions. For example, a hospital must protect patient privacy, but it must also keep life-critical systems available. A bank must protect transaction accuracy while ensuring customers can still access services when needed.

Together, these goals turn cybersecurity into a business, institutional, and national reliability issue—not only a technical one.

4. Cybersecurity in Daily Life

Cybersecurity affects ordinary users every day. It helps protect online banking, digital wallets, shopping accounts, email, messaging apps, cloud storage, and social media profiles. Many daily security measures—such as fingerprint unlock, facial recognition, password managers, software updates, and two-factor authentication—are practical expressions of cybersecurity.

It also supports connected environments such as smart homes, wearable health devices, remote learning, and hybrid work. When these systems are poorly protected, users can face account theft, fraud, stalking, blackmail, privacy loss, or service disruption.

For this reason, public cyber guidance often focuses on habits as much as on technology: choosing strong passwords, enabling multi-factor authentication, updating devices, recognizing phishing, and backing up important files.

5. Common Cyber Threats

Phishing is one of the most common threats. Attackers send fake emails, texts, or websites to trick users into revealing passwords, financial details, or one-time codes. The attack succeeds by exploiting trust rather than by breaking software directly.

Malware refers to malicious software designed to damage systems, spy on users, steal data, or give attackers unauthorized control. Ransomware is a specific type of malware that encrypts files or systems and demands payment for restoration. It can stop business operations, delay healthcare services, and harm public institutions.

Social engineering is broader than phishing and includes all methods that manipulate human behavior. Other major threats include distributed denial-of-service attacks, which overload services until they become unavailable, and supply-chain attacks, which spread compromise through trusted software, vendors, or service providers.

6. Main Categories of Cybersecurity

Network security focuses on protecting internal and external communications through technologies and practices such as segmentation, firewalls, intrusion prevention, and secure configurations. It aims to reduce unauthorized movement inside networks and limit exposure to the internet.

Application security protects websites, business software, and mobile apps from weaknesses such as insecure coding, poor authentication, or unpatched vulnerabilities. Data security protects files, records, databases, and sensitive content through encryption, classification, backups, and controlled access.

Cloud security addresses workloads and data stored in platforms such as Microsoft 365, Google Cloud, AWS, or Azure. Endpoint security protects laptops, desktops, phones, servers, and other user-facing devices that often serve as the first entry point in an attack.

7. Specialized Branches and Operations

Identity and Access Management determines who can access which systems, under what conditions, and with what privileges. It includes account lifecycle control, least privilege, single sign-on, and strong authentication. Because attackers often aim to steal credentials, identity security has become central to modern defense.

IoT and OT security protect connected devices, industrial systems, smart sensors, utilities, and operational technology environments such as energy grids, factories, and transportation systems. These systems are especially important because failures can affect public safety and essential services.

Security Operations Centers monitor alerts, logs, and indicators of suspicious activity. Threat intelligence helps organizations understand attacker behavior, techniques, and campaigns. Incident response and digital forensics manage containment, investigation, evidence preservation, and recovery. Governance, Risk, and Compliance connects all of this work to policy, legal requirements, standards, and executive decision-making.

8. How Cybersecurity Works in Practice

A practical cybersecurity program begins by identifying assets, risks, and priorities. Organizations need to know what they are protecting, why it matters, and what harm would result from compromise. This includes systems, people, suppliers, data, and critical processes.

After identification comes layered protection. Typical controls include access control, encryption, patching, secure configuration, employee training, backups, network segmentation, and endpoint defenses. Good security does not depend on a single product; it relies on multiple reinforcing layers.

Monitoring and detection follow. Teams watch logs, alerts, and behavioral signals to detect anomalies before they become larger incidents. When incidents occur, organizations need response plans, communication procedures, restoration priorities, and lessons-learned reviews. This is why cybersecurity is best understood as a continuous cycle rather than a one-time setup.

9. Cybersecurity Careers

Cybersecurity offers a wide range of careers. SOC analysts watch security alerts and investigate suspicious activity. Penetration testers or ethical hackers simulate attacker behavior to discover weaknesses before criminals do. Security engineers design and build secure systems.

Incident responders and digital forensics specialists handle breaches, preserve evidence, and reconstruct what happened. Cloud security engineers focus on modern platforms and shared-responsibility models. GRC analysts align operations with standards, law, and policy. Senior leaders such as CISOs connect security strategy to business risk and executive decision-making.

Importantly, cybersecurity is not limited to pure programming. It includes communication, risk analysis, policy, training, compliance, architecture, law, audit, and leadership. This makes the field attractive to students from different backgrounds.

10. Impact on Society and Countries

At the social level, cybersecurity protects privacy, identity, trust, and confidence in digital systems. Without it, online education, e-commerce, digital health, and remote work become fragile. Citizens may become less willing to adopt services if they believe fraud, surveillance, or account theft are unavoidable.

At the national level, cybersecurity supports resilience in critical infrastructure such as power, water, telecommunications, healthcare, transportation, finance, and government services. If these sectors are disrupted, the consequences can affect safety, economic stability, and public trust.

As a result, cybersecurity has become part of national security and state capacity. A country's cyber strength influences resilience, deterrence, public service continuity, intelligence protection, and strategic competitiveness.

11. Cyber Operations, Competition, and Ethics

Cyber capabilities can be used for defense, but they can also be used for espionage, disruption, coercion, sabotage, and influence operations. State and state-sponsored groups may target government networks, critical infrastructure, or strategic industries to collect intelligence or prepare future options in times of tension.

This is why cybersecurity is often described as a double-edged weapon. The same technical knowledge used to secure a system can also be adapted to exploit weaknesses. Strong cyber capabilities may deter attacks, but they can also intensify strategic competition if not constrained by law, ethics, and international norms.

For that reason, modern cybersecurity must include governance, accountability, resilience, and responsible behavior. Technical strength matters, but so do policy, legal frameworks, public awareness, and institutional preparedness.

12. Final Takeaways

Cybersecurity is no longer just a support function inside IT departments. It shapes personal safety, business continuity, social trust, and national resilience. In a world where value is increasingly digital, security is part of the foundation of modern life.

Every user is part of the defense chain. Good security depends on both systems and people: safe behavior, informed decisions, strong identity protection, regular updates, backups, and rapid reporting when something looks suspicious.

For organizations and countries, the lesson is clear: cybersecurity is not optional, and it is not finished once tools are installed. It requires continuous governance, investment, practice, and improvement.

Quick Summary Table

Area	Why It Matters	Examples
Personal Life	Protects privacy, money, and accounts	Banking, email, MFA, cloud storage
Organizations	Protects operations and trust	Backups, monitoring, incident response
Countries	Protects critical services and resilience	Power, telecom, healthcare, government
Conflict & Competition	Can be used for espionage or disruption	Pre-positioning, credential theft, influence ops

13. References

CISA. “What is Cybersecurity?”

ENISA. “Threat Landscape” and “ENISA Threat Landscape 2024.”

NIST. “Cybersecurity Framework” and “The NIST Cybersecurity Framework (CSF) 2.0.”

NICCS/CISA. “NICE Workforce Framework for Cybersecurity.”

NCSC (UK). “Cyber Aware,” “Shopping and paying safely online,” “Keeping devices and software up to date,” and MFA guidance.

CISA. “Nation-State Threats,” Iran threat advisories, and PRC state-sponsored actor advisory AA24-038A.

NATO. “Cyber defence” topic pages and the July 10, 2024 announcement on the NATO Integrated Cyber Defence Centre.

https://www.cisa.gov/news-events/news/what-cybersecurity

https://www.enisa.europa.eu/topics/cyber-threats/threat-landscape

https://www.enisa.europa.eu/publications/enisa-threat-landscape-2024

https://www.nist.gov/cyberframework

https://www.nist.gov/publications/nist-cybersecurity-framework-csf-20

https://niccs.cisa.gov/tools/nice-framework

https://www.ncsc.gov.uk/cyberaware/home

https://www.ncsc.gov.uk/guidance/shopping-online-securely

https://www.ncsc.gov.uk/collection/device-security-guidance/managing-deployed-devices/keeping-devices-and-software-up-to-date

https://www.ncsc.gov.uk/collection/mfa-for-your-corporate-online-services/recommended-types-of-mfa

https://www.cisa.gov/topics/cyber-threats-and-advisories/nation-state-cyber-actors

https://www.cisa.gov/news-events/cybersecurity-advisories/aa24-290a

https://www.cisa.gov/news-events/cybersecurity-advisories/aa24-038a

https://www.nato.int/en/what-we-do/deterrence-and-defence/cyber-defence

https://www.nato.int/en/news-and-events/articles/news/2024/07/10/allies-agree-new-nato-integrated-cyber-defence-centre